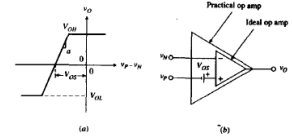


Opamp performance parameters

ARN

Input offset Voltage Vos

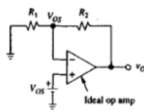
- $V_o = a(V_p - V_n) = 0$ @ $V_p = V_n$
- Due to mismatch in the i/p transistor practical opamp produce nonzero V_o
- V_{os} the input offset voltage needed at input so as $V_o = 0V$
- $V_o = a[V_p + V_{os} - V_n]$



IC	741	741E	Op07	op77
Vos (typical)	2mv	0.8mv	30uV	10uV
Vos (Max)	6mv	3mv	75uV	50uV

- In Figure below the offset-free op amp acts as a non-inverting amplifier with respect to V_{os} . so $V_o = E_o$. Where E_o is the output error

$$E_o = \left(1 + \frac{R_2}{R_1}\right) V_{os}$$



- For instance, with $R_1 = R_2$. a 741C op amp yields $E_o = (1 + 1) \times (\pm 2 \text{ mV}) = \pm 4 \text{ mV}$ typical, $(1 + 1) \times (\pm 6 \text{ mV}) = \pm 12 \text{ mV}$ maximum.
- However with $R_2 = 1000 \times R_1$ it yields $E_o = (1 + 1K) \times (\pm 2 \text{ mV}) = \pm 2V$ typically.
- $\pm 6 \text{ V}$ maximum -quite an error!

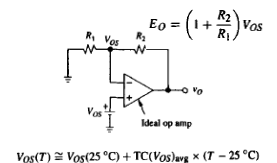
Error caused due to Vos

- Thermal drift: V_{os} is temperature dependent

$$T_c(V_{os}) = \frac{\partial V_{os}}{\partial T} = \mu V/oC$$

- Compute V_o @ 70oC

- For instance an opamp with $V_{os}(@25^\circ C) = 1 \text{ mV}$ and $T_c(V_{os})_{avg} = 5 \mu V/oC$
- would have $V_{os}(70^\circ C) = 1 \text{ mV} + (5 \mu V/oC) \times (70 - 25) = 1.225 \text{ mV}$.

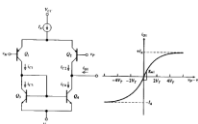


$$V_{os}(T) \cong V_{os}(25^\circ C) + T_c(V_{os})_{avg} \times (T - 25^\circ C)$$

IC	741	op77
Tc(Vos) (typical)	5uV/oC	0.1uV/oC
Tc(Vos) (Max)		0.3uV/oC

Input Bias current, input offset current

- Practical Opamp draw small current in their input pins
- The I_p and I_n are needed to bias the i/p transistor of diff-amp
- These currents are derived from external circuit
- Due to mismatch in the β of input stage $I_p \neq I_n$
- The average current $(I_p + I_n)/2$ is called input bias current I_B
- The difference between I_p and I_n is called input offset current $I_{os} = I_p - I_n$



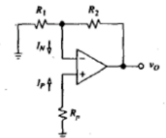
Error caused by input bias/ offset current

IC	741	741E	Op77
IB(typ)	80nA	30nA	1.2nA
IB max	500nA	80nA	2nA
Ios(typ)	20nA	3nA	0.3nA
Ios(max)	200nA	30nA	1.5nA

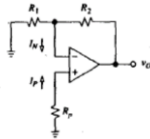
$$V_P = -R_P I_P$$

$$V_O = (1 + R_2/R_1) V_P + R_2 I_N = R_2 I_N - (1 + R_2/R_1) R_P I_P$$

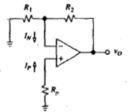
By using super position theorem, set I_p zero compute V_{on} due to I_n . Set I_n zero compute V_{op} due to I_p . The $V_o = V_{op} + V_{on}$



- Assume $I_n=0$, $V_{op} = (1+R_2/R_1) * (-I_p R_p)$
- $V_{on} = R_2 * I_n$
- $V_o = V_{op} + V_{on} = R_2 * I_n - I_p R_p (1 + R_2/R_1)$
- $V_o = (1+R_2/R_1) [(R_1 R_2 / R_1 + R_2) I_n - I_p R_p]$
- $V_o = (1+R_2/R_1) [(R_1 || R_2) I_n - I_p R_p]$
- $V_o = (1+R_2/R_1) [(I_n - I_p) * R_p]$, if $R_p = R_1 || R_2$
- $V_o = (1+R_2/R_1) [I_{os} * R_p]$
- $V_o = (R_1 + R_2) / R_1 * R_1 R_2 / R_1 + R_2 * I_{os} = R_2 * I_{os}$



- $V_o = (1+R_2/R_1) [(R_1 || R_2) I_n - R_p I_p]$
- Let $C = (1+R_2/R_1)$
- Let $B = R_p$
- Let $A = R_1 || R_2$
- $V_o = C [A I_n - B I_p]$
- $V_o = C [2 A I_n - 2 B I_p] / 2$
- $V_o = C [A I_n + A I_n - B I_p - B I_p] / 2$
- $V_o = C [A I_p + A I_n - B I_p - B I_n - A I_p + A I_n - B I_p + B I_n] / 2$
- $V_o = A (I_p + I_n) / 2 - B (I_p + I_n) / 2$
- $V_o = C [A (I_B) - B (I_B) - A (I_{os}/2) - B (I_{os}/2)]$
- $V_o = C [A - B] I_B - [A + B] I_{os} / 2$
- $V_o = (1+R_2/R_1) [(R_1 || R_2) - R_p] I_B - [(R_1 || R_2) + R_p] I_{os} / 2$
- If $R_1 || R_2 = R_p$
- Then $V_o = - [(R_1 || R_2) + R_p] I_{os} / 2 * (1+R_2/R_1)$
- $V_o = - [(R_1 || R_2) * 2] I_{os} / 2 * (1+R_2/R_1)$
- $V_o = - (R_1 || R_2) I_{os} * (1+R_2/R_1)$
- $V_o = R_1 R_2 / R_1 + R_2 * R_1 + R_2 / R_1 * I_{os}$
- $V_o = R_2 * I_{os}$

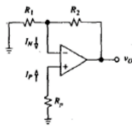


Example: let $R_1 = 22 \text{ k}\Omega$ and $R_2 = 2.2 \text{ M}\Omega$, and let the op amp ratings be $I_B = 80 \text{ nA}$ and $I_{os} = 20 \text{ nA}$.

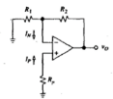
- Calculate E_o for the case $R_p = 0$.
- Repeat, but with $R_p = R_1 || R_2$ in place.
- Repeat part (b), but with all resistances simultaneously reduced by a factor of 10.
- Repeat part c, but with the op amp replaced by one with $I_{os} = 3 \text{ nA}$. Comment.

$$E_o = (1+R_2/R_1) [(R_1 || R_2) - R_p] I_B - [(R_1 || R_2) + R_p] I_{os} / 2$$

- 175mV
- 44mV
- 4.mV
- 0.7mV



Measuring input offset voltage



- we can use the present circuit to measure V_{os} .
- For instance, let $R_1 = 10 \Omega$ and $R_2 = 10 \text{ k}\Omega$. so that the DC noise gain is 1001 V/V and the combination $R_1 || R_2$ is sufficiently small to make the effect of I_n negligible.
- Suppose we measure the output and find $E_o = -0.5 \text{ V}$.
- Then $V_{os} = E_o / 1001 = -0.5 \text{ mV}$. a negative offset in this specific example

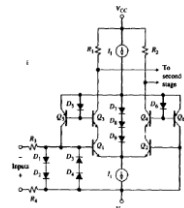
Low input bias current opamp (Super β i/p)

- One way of achieving low I_{os} is by using input BJTs with extremely high current gains β in excess of 10^3 . Known as super-beta transistors
- This technique was pioneered with the LM308 op amp (National Semiconductor).
- Super-beta op amps have typically $I_B \sim 1 \text{ nA}$ or less.

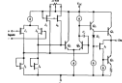
Parameter	Condition	LM108/LM208			LM308			Units
		Min	Typ	Max	Min	Typ	Max	
Input Offset Voltage	$T_A = 25^\circ\text{C}$		0.7	2.0		2.0	7.5	mV
Input Offset Current	$T_A = 25^\circ\text{C}$		0.05	0.2		0.2	1	nA
Input Bias Current	$T_A = 25^\circ\text{C}$		0.8	2.0		1.5	7	nA
Input Resistance	$T_A = 25^\circ\text{C}$	30	70		10	40		M Ω
Supply Current	$T_A = 25^\circ\text{C}$		0.3	0.6		0.3	0.8	mA
Large Signal Voltage Gain	$T_A = 25^\circ\text{C}$, $V_{IS} = \pm 15 \text{ V}$, $V_{OUT} = \pm 10 \text{ V}$, $R_L \geq 10 \text{ k}\Omega$	50	300		25	300		V/mV

Input-Bias-Current Cancellation

- Another popular technique for achieving low I_B is current cancellation
- A Special circuitry anticipates the base currents needed to bias the input transistors, then itself supplies these currents internally, making the op amp appear to an outsider as if it were capable of operating without any input bias current.
- The ratings of OP-07 are $I_B = \pm 1 \text{ nA}$ and $I_{OS} = 0.4 \text{ nA}$.

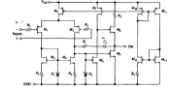


JFET-Input Op Amps



- These devices realize the input-stage differential pair with junction field-effect transistors (JFETs), and the remaining circuitry with conventional BJTs.
- Now I_B is the JFET gate current, which is the reverse bias current of the pn junction between gate and channel.
- At room temperature this current is typically on the order of a few tens of pico amperes or less.
- The room-temperature ratings for the LF356 are $I_B = 30$ pA and $I_{os} = 3$ pA.
- The AD549 (Analog Devices) and OPA129 (Burr-Brown) op amps use special JFET structures and isolation techniques to achieve $I_B < 100$ fA.
- These devices find application in electrometer, ion gauge, and photodetector amplifiers.

MOSFET-Input Op Amps



- When the differential input pair is implemented with metal-oxide-silicon FETs (MOSFETs).
- I_B is the leakage current of the gate-channel capacitor.
- This current is typically in the range of a few pico amperes.
- In BiMOS op amps the input pair is in MOS technology and the rest of the circuitry in bipolar.
- Typical room-temperature ratings are $I_B = 0.7$ pA and $I_{os} = 0.1$ pA, for IC TLC279 of TI

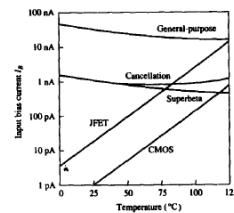
Input-Bias-Current Drift

- We observe that in BJT-input devices I_B tends to decrease with temperature, owing to the fact that β increases with temperature
- for JFET-input devices, I_B increases exponentially with temperature.
 - A well-known rule of thumb states that the reverse-bias current of a pn junction, whether it is that of a diode or of a JFET, doubles for every 10°C increase.
 - Once we know I_B at some reference temperature T_0 , we can predict it at any other temperature T by

$$I_B(T) \cong I_B(T_0) \times 2^{(T-T_0)/10}$$

A certain FET-input op amp is rated $I_B = 1$ pA at 25°C .
Estimate I_B at 100°C .
ANS $I_B(100^\circ\text{C}) = 1 \times 10^{-12} \times 2^{(100-25)/10} = 0.18$ nA

Input-Bias-Current Drift



CMRR

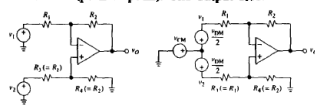
- Opamp responds to differential input ie $V_o = a(V_p - V_n)$
- Practical opamps are sensitive to common mode input voltage $V_{cm} = (V_p + V_n)/2$
- The transfer characteristics of opamp can be represented by
 - $V_o = a(V_p - V_n) + a_{cm}V_{cm}$
 - $V_o = a[V_p + (a_{cm}/a)V_{cm} - V_n]$
 - $V_o = a[V_p + (V_{cm}/CMRR) - V_n]$
- The common mode voltage alter operating point of internal transistors and produces error in output voltage

- We know $V_o = a(V_p - V_n)$
- $A_d/A_{cm} = dV_o/d(V_p - V_n) \cdot d(V_p - V_n)/dV_o$
- $A_d/A_{cm} = dV_o/dV_{cm} \cdot dV_{cm}/d(V_p - V_n)$,if $V_p = V_n$
- $1/CMRR = dV_o/dV_{cm}$ and can be specified as $\mu\text{Volts/Volt}$
- CMRR is specified in dB
- $CMRR_{dB} = 20 \log CMRR$
- $1/CMRR = 10^{-CMRR_{dB}/20}$
- CMRR is frequency dependent and start roll-off at 100Hz.
- CMRR is no serious concern for inverting amplifier as V_p is at 0 volts

EXAMPLE 5A. The difference amplifier of Fig. 2.13 uses a 741 op amp and a perfectly matched resistance set with $R_1 = 10 \text{ k}\Omega$ and $R_2 = 100 \text{ k}\Omega$. Suppose the inputs are tied together and driven with a common signal v_I . Estimate the typical change in v_O if (a) v_I is slowly changed from 0 to 10 V, and (b) v_I is a 10-kHz, 10-V peak-to-peak sine wave.

Solution.

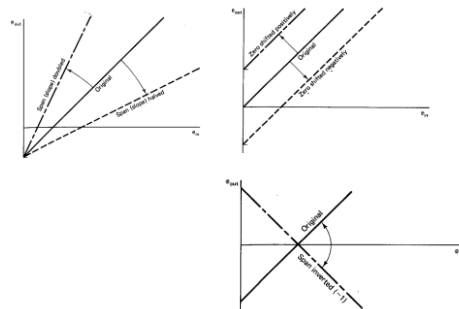
- (a) At dc we have $1/\text{CMRR} = 10^{-90/20} = 31.6 \text{ }\mu\text{V/V}$, typical. The common-mode change at the op amp input pins is $\Delta v_P = [R_2/(R_1 + R_2)]\Delta v_I = [100/(10 + 100)]10 = 9.09 \text{ V}$. Thus, $\Delta v_{OS} = (1/\text{CMRR})\Delta v_P = 31.6 \times 9.09 = 287 \text{ }\mu\text{V}$. The dc noise gain is $1 + R_2/R_1 = 11 \text{ V/V}$. Hence, $\Delta v_O = 11 \times 287 = 3.16 \text{ mV}$.
- (b) From the CMRR curve of Fig. 5A.6 we find $\text{CMRR}_{\text{dB}}(10 \text{ kHz}) \cong 57 \text{ dB}$. So, $1/\text{CMRR} = 10^{-57/20} = 1.41 \text{ mV/V}$, $\Delta v_{OS} = 1.41 \times 9.09 = 12.8 \text{ mV}$ (peak-to-peak), and $\Delta v_O = 11 \times 12.8 = 0.141 \text{ V}$ (peak-to-peak). The output error at 10 kHz is much worse than at dc.



Zero and Span circuit

Need...

- The output of transducer may not match with ADC requirement
- The zero (intercept on Y axis, c) and span (slope, m) circuit allows you to produce any convenient parameter (Temp. pressure etc.) to voltage relationship



Example

When the temperature in a process is at its minimum, the sensor outputs 2.48 V. At maximum temperature, it outputs 3.90 V. The A/D converter used to input these data into a computer has the range 0 to 5 V. To provide maximum resolution, you must zero and span the signal from the transducer so that it fills the entire range of the converter.

